ELEC70091
EE9-CS2-1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Wednesday, 14 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the following nonlinear dynamical system:

$$\begin{aligned} \dot{x}_1 &= -x_1(x_1^2-1)(x_1^2-4)+x_2 \\ \dot{x}_2 &= -x_2-2x_1. \end{aligned}$$

Let $x=[x_1,x_2]'$ denote the state of the system, evolving in $\mathbb{R}^2$.

a) Discuss existence and uniqueness of solutions. [2]

b) Describe analytically and sketch graphically the nullclines of the vector-field (Hint: find intersections of the first nullcline with the x_1 axis for a better sketch). [2]

c) Find regions in $\mathbb{R}^2$ delimited by nullclines and identify the orientation of the vector-field in each one of them. [4]

d) Find all forward invariant regions among those identified in part c) (Justify your answer). [2]

e) Compute all equilibria of the system (Hint: after eliminating the trivial solution, solve the resulting polynomial equation using x_1^2 as your unknown). [3]

f) Linearize the system around each equilibrium and discuss its local phase portrait. [4]

g) Sketch a global phase-portrait for the system. [3]

2. Consider the following nonlinear second-order differential equation:

$$\ddot{z}(t) = -\dot{z}(t)z^2(t) - \dot{z}^3(t) - \sin(z(t)) + u(t),$$

where $z(t)$ and $u(t) \in \mathbb{R}$.

a) Choose a suitable state variable x and write the system's equations in the corresponding state-space representation, with input $u(t)$ and output $y(t) := \dot{z}(t)$. [4]

b) Is the system input-affine ? (Justify your answer). [2]

c) Show that the system is passive (from u to y) by finding an appropriate storage function $S(x)$. [4]

d) Show that the equilibrium corresponding to $u = 0$ and $z = \pi$ is unstable. [4]

e) Consider next the feedback $u = -z$. Prove that the resulting closed-loop system is Globally Asymptotically Stable in $z = 0$, $\dot{z} = 0$, using a suitable candidate Lyapunov function and stability criterion (Hint: use as a candidate Lyapunov function a modified storage function, from part c)). [6]

3. Consider the following nonlinear system:

$$\begin{aligned} \dot{x}_1 &= -x_1^3 + x_2 \\ \dot{x}_2 &= -x_2^3 + a x_1^9 \end{aligned}$$

with state $x = [x_1, x_2]^T$ and $a \in \mathbb{R}$ an uncertain parameter.

a) Rewrite the system as a feedback interconnection of 2 nonlinear scalar input-output systems Σ_1, Σ_2. [4]

b) Show that each subsystem is Input-to-State Stable with respect to its input signal, and compute the associated $\mathscr{K}_\infty$ gains γ_1, γ_2. (Hint: Try to be as tight as possible in estimating the gains). [6]

c) Consider the cascade interconnection of Σ_1 and Σ_2. Show that the cascade is ISS and compute the associated gain from its input to its output, taken to be the input of the upstream system and the state of the downstream one respectively (Hint: use the notion of Asymptotic Gain, to prove a suitable inequality between input and output signal norms). [3]

d) Consider next the feedback interconnection as in part a). Show that for $|a| < 1$ a small gain condition holds and prove that the origin is globally asymptotically stable. [3]

e) Consider next the Σ_1 subsystem in isolation and assume that it is forced by an exogenous input $d_1(t) = (1 + e^{-t})\cos(t)$. Find an upperbound to the asymptotic amplitude of the corresponding solution, $\limsup_{t \to +\infty} |x_1(t)|$ (from an arbitrary initial condition $x_1(0)$). [4]

4. Consider the nonlinear control system of equations:

$$\begin{aligned} \dot{x}_1 &= x_3 - (x_1 - x_2)^3 + x_1 + x_2 \\ \dot{x}_2 &= x_3 + (x_1 - x_2)^3 - x_1 - x_2 \\ \dot{x}_3 &= (1 + x_1^2)^{-1} u + \cos(x_2) \\ \\ y &= x_1 + x_2 \end{aligned}$$

with state $x = [x_1, x_2, x_3]' \in \mathbb{R}^3$, and scalar input and output variables u and y, respectively.

a) Compute the relative degree of the system and justify whether this is locally or globally defined. [2]

b) Find a state-feedback $u = k(x, v)$ so that the resulting closed-loop system is linear from the auxiliary input $v \in \mathbb{R}$ to output y. [3]

c) Design a stabilizing linear feedback control action $v = k_1 y + k_2 \dot{y}$ so that the Input-Output dynamics and the equilibrium corresponding to $y(t) \equiv 0$ are globally asymptotically stabilized in closed-loop. [3]

d) Bring the closed-loop system in normal form and highlight the internal dynamics. [4]

e) Assess whether the resulting internal dynamics are Input-to-State Stable (ISS). [4]

f) Comment on the implications of the previous ISS analysis on the stability of the overall system. [4]